Homework 7

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1 Notes on Video Lectures

1.1 Video 1

Capital budgeting was a Theory of Interest topic but is more complicated with corporate finance theory.

Idea: how do we decide whether to undertake a project? Answer: need idea of expected return. This requires adding up cashflows. Question: which ones count and which ones don't count? We only consider cashflows from assets; financing will eventually matter but we consider it here as part of the weighted average cost of capital (WACC).

Which cashflows count? Say Coca-Cola develops new product, Coke Infinity. Include . . .

1. Revenue from sales of *Coke Infinity*: yes
2. Cost of production: yes
3. existing factory: ~~no~~ yes? because it would be a future cashflow
4. marketing of new product: ~~no~~ yes – difference between this and the next two is that the latter would be taken whether or not the new product is introduced; they represent costs already paid (sunk costs)
5. cost of focus group: no
6. research & development: no
7. taxes: yes
8. reduced consumption of other coke products? ~~no~~ can't ignore effect (does it matter

that they can't be projected easily??)

Also should consider: increase in net working capital, NWA (could be temporary!); also depreciating fixed assets to reduce taxes could count in here.

project cash flow:

(sales - costs - depreciation) - taxes + depreciation - (capital spending + change in NWC) = project cash flow Also intermediate values! EBIT (earnings before interest & taxes): include up to taxes. Net income: include taxes. Operating cash flow: include depreciation.

NPV of project cash flows: $NPV = \sum_{t \geqslant 0} CF_t(1+i)^{-t}$. i represents the WACC and will be discussed later. Obviously the firm wants to undertake the project if the NPV is positive.

Depreciation only affects CF through taxes. If project requires initial investment X, then it should get a tax credit τX. US laws says tax cannot be taken all at once if the project is taken over several years; we calculate this using depreciation of the book value of the asset. (easy: divide the total tax credit over the length of the project.) n.b.: because of time value of money, firms would rather not do this!

Modified Accelerated Cost Recovery System (MACRS): faster depreciation realizes tax credit earlier, though it also assumes things are bought in the middle of the year rather than the beginning. (is this actually used? the previous discussion said US law requires the other schedule.)

1.2 Video 2

Project has initial cost of $\$500,000$ and is also the amount depreciated using a three-year MACRS schedule.

Year	Depreciation rate	Depreciation	Book value at year end
0			500000
1	33.33%	166650	333350
2	44.45%	222250	111100
3	14.81%	74050	37050
4	7.41%	37050	0

Capital spending: spend \$500000 in year 0, sell assset and receive salvage value of \$50000 in year 4. less 40% in taxes gives \$30000.

change to NWC: spend \$40000 at 0, get money back at 4.

Pro-forma income & CF statement:

Year	0	1	2	3	4
Sales		600000	600000	600000	600000
- Costs		400000	400000	400000	400000
- Depreciation		166650	222250	74050	(37050)
EBIT		33350	(22250)	125950	162950
- taxes		(13340)	8900	(50380)	(65180)
Net Income		20010	(13350)	75570	97750
+ Depreciation		166650	222250	74050	(37050)
Operating CF		186660	208900	149620	134820
- capital spending	(500000)	0	0	0	30000
- change in NWC	(40000)	0	0	0	40000
Project CF	(540000)	−186660	208900	149620	204820

NPV: $-540000 + \frac{186660}{1.10} + \frac{208900}{1.10^2} + \frac{149620}{1.10^3} + \frac{204820}{1.10^4} = 54642.07$. This value is positive, so the firm should undertake the project.

1.3 Video 3

Start with a depreciation table:

Year	Depreciation (old eq.)	BV (old eq.)	Depreciation (new eq.	BV (new eq.))
0		250000		760000
1	50000	200000	152000	608000
2	50000	150000	152000	456000
3	50000	100000	152000	304000
4	50000	50000	152000	152000
5	50000	0	152000	0

so depreciation increases by 102000 each year.

Capital spending: spend 760000 at 0. sell old equipment, minus taxes: $200000 \times (1 - 30\%) = 140000$. Firm is entitled to tax credit of $30\% \times$ book value $= 30\% \times 250000 = 75000$. So total capital spending is -545000. At time 5: sell new equipment, minus taxes: $150000 \times (1 - 30\%) = 105000$.

Incremental income/CF statement can now be generated.

Year	0	1-4	5
Sales - costs		200000	200000
- Depreciation		(102000)	(102000)
EBIT		98000	98000
- taxes		(29400)	(29400)
Net Income		68600	68600
+ Depreciation		102000	102000
Operating CF		170600	170600
- capital spending	(545000)	0	105000
- change in NWC	(20000)	0	20000
Project CF	(565000)	170600	295600

NPV: $-565000 + \frac{170600}{1.15} + \cdots + \frac{170600}{1.15^4} + \frac{295000}{1.15^5} \approx 69025 > 0$ so replace the equipment.

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The depreciation schedule is created entirely ad-hoc by an act of law, so there is no need to calculate depreciation. Thus we skip straight to building the table.

Year	0	1	2	3
Sales - costs		30000	30000	30000
- Depreciation		(40000)	(5000)	(5000)
EBIT		(10000)	25000	25000
- taxes		0	10000	10000
Net Income		(10000)	15000	15000
+ Depreciation		40000	5000	5000
Operating CF		30000	20000	20000
- capital spending	(50000)	0	0	10000
- change in NWC	(12000)	0	0	12000
Project CF	(62000)	30000	20000	42000

Based on this analysis, the net present value is given by

$$\text{NPV} = -62000 + \frac{30000}{1.15} + \frac{20000}{1.15^2} + \frac{42000}{1.15^3} = 6825.51$$

The net present value is positive, so the project is worth the investment.